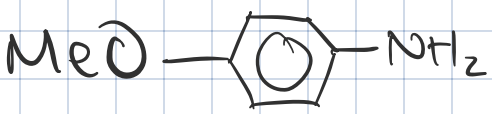
| | pKb |
|---|---|
| $NH_3$ | 4.7 |
| $MeNH_2$ | 3.4 |
| $Me_2NH$ | 3.3 |
| $Me_3N$ | 4.3 |
| $Ph-NH_2$ | 9.4 |
| 4-nitroaniline | 13 |
| p-Anisidine  | 8.7 |

Review: explain all of these

(answers on next page)

$MeNH_2$: More basic than $NH_3$, because conjugate acid is stabilized by methyl's hyper conjugation

3.4

$Me_2NH$: Diminishing returns on hyperconjugation versus conjugate acid's low solvation in water ($CH_3$ groups bulky/nonpolar)

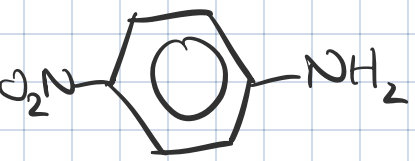
3.3

$Me_3N$: Hyperconjugation makes it still stronger than $NH_3$, but significantly weaker than the other methylamines be of low solvation enthalpy

4.3

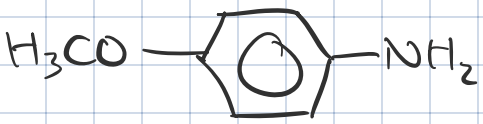
: 1 major resonance structure is basic. The other 3, which are minor but still effective, delocalize the negative charge to the ring, thus significantly lowering the basicity.

9.4



13

: Similar to Ph–Am, but now we have a strong EWG in para, pulling hard on the basic lone pair on the am!



: Similar to Ph–Am, but EDG makes lp less distributed towards the ring and more on the $NH_2$ itself.